

STRUCTURAL WIND LOAD CALCULATIONS

WAREHOUSE EXPANSION — PHASE II | Job #2024-W-0147 | March 12, 2026

Global Wind Design Criteria

Parameter	Value	Parameter	Value
Building Code	7-22	Exposure Category	C
Wind Speed (V)	120 mph	Risk Category	II
Enclosure	Enclosed	GC _{pi}	±0.18
Length (L)	120 ft	Width (B)	80 ft
Mean Roof Ht (h)	35 ft	Roof Angle (θ)	4.0°
K _e	0.982063	K _{zt}	1.0000
K _d	0.85	G (Gust Factor)	0.8682

Velocity Pressure Determination

ASCE 7-22 §26.10

Ground Elevation Factor

ASCE 7 §26.9, Eq. 26.9-1

$$K_e = e^{-0.0000362 z_g}$$
$$= e^{-0.0000362 \times 500}$$

K_e = 0.982063

Velocity Pressure Exposure Coefficient at z = 35 ft

ASCE 7 §26.10.1, Table 26.10-1

$$K_z = 2.01 (z / z_g)^{2/\alpha}$$
$$= 2.01 \times (35.0 / 900)^{2/9.8}$$

K_z = 1.036109

Velocity Pressure at z = 35 ft

ASCE 7 §26.10, Eq. 26.10-1

$$q_z = 0.00256 K_e K_z K_{zt} K_d V^2$$
$$= 0.00256 \times 0.9821 \times 1.0361 \times 1.0000 \times 0.85 \times 120^2$$

q_z = 31.8835 psf

Velocity Pressure Profile

z (ft)	K _z	K _{zt}	K _e	q _z (psf)
15.0	0.8716	1.0000	0.9821	26.82
20.0	0.9243	1.0000	0.9821	28.44
25.0	0.9673	1.0000	0.9821	29.77
30.0	1.0040	1.0000	0.9821	30.90
35.0	1.0361	1.0000	0.9821	31.88

MWFRS Wall Pressure Coefficients

ASCE 7-22 §27.3

Leeward Wall External Pressure Coefficient

ASCE 7 Figure 27.3-1

$$C_{p, \text{leeward}} = f(L/B) \text{ from Figure 27.3-1}$$

$$L/B = 1.500, \text{ interpolating between } 1.0 (C_p = -0.50) \text{ and } 2.0 (C_p = -0.30)$$

$$C_{p, \text{leeward}} = -0.4000$$

Leeward Wall Design Pressure

ASCE 7 §27.3, Eq. 27.3-1

$$p = q_h G C_p - q_i (\pm GC_{pi})$$

$$= 31.88 \times 0.87 \times (-0.4000) \pm 31.88 \times 0.18$$

$$p = -5.33 \text{ psf (with } +GC_{pi}) / -16.81 \text{ psf (with } -GC_{pi})$$

MWFRS — Directional Procedure (Chapter 27)

ASCE 7 §27.3, Eq. 27.3-1

Surface	C_p	$p (+GC_{pi})$	$p (-GC_{pi})$
Windward Wall	0.80	varies by z	varies by z
Leeward Wall	-0.4000	-5.33 psf	-16.81 psf
Side Wall	-0.70	-13.64 psf	-25.12 psf

Parapet Pressures (§26.10.3): Windward = **47.83 psf**, Leeward = **-31.88 psf**

Windward Wall Pressure Profile

z (ft)	K_z	q_z (psf)	$p (+GC_{pi})$	$p (-GC_{pi})$
15.0	0.8716	26.82	24.37	12.89
20.0	0.9243	28.44	25.49	14.02
25.0	0.9673	29.77	26.42	14.94
30.0	1.0040	30.90	27.20	15.72
35.0	1.0361	31.88	27.88	16.41

MWFRS — Low-Rise Envelope Procedure (Chapter 28)

ASCE 7 §28.3, Figure 28.3-1

 $q_h = 31.88 \text{ psf}$ | End zone width (2a) = 16.0 ft

Case A — Transverse

Zone	GC_{pf}	$p (+GC_{pi})$ psf	$p (-GC_{pi})$ psf
1	0.4260	19.32	7.84
2	-0.6900	-16.26	-27.74
3	-0.3920	-6.76	-18.24
4	-0.3180	-4.40	-15.88
1E	0.6480	26.40	14.92
2E	-1.0700	-28.38	-39.85
3E	-0.5620	-12.18	-23.66
4E	-0.4720	-9.31	-20.79

Case B — Longitudinal

Zone	GC_{pf}	$p (+GC_{pi})$ psf	$p (-GC_{pi})$ psf
1	-0.4500	-8.61	-20.09
2	-0.6900	-16.26	-27.74

Zone	GC _{pf}	p (+GCpi) psf	p (-GCpi) psf
3	-0.3700	-6.06	-17.54
4	-0.4500	-8.61	-20.09
5	0.4000	18.49	7.01
6	-0.2900	-3.51	-14.99
1E	-0.4800	-9.57	-21.04
2E	-1.0700	-28.38	-39.85
3E	-0.5300	-11.16	-22.64
4E	-0.4800	-9.57	-21.04
5E	0.6100	25.19	13.71
6E	-0.4300	-7.97	-19.45

Components & Cladding — h ≤ 60 ft (§30.3)

ASCE 7 Figure 30.3-1 | Roof angle range: 0° to 7°

q_h = 31.88 psf | GC_{pi} = ±0.18 | Zone dimension a = 8.0 ft | Minimum pressure = 16.0 psf**GC_p Coefficient Determination**

ASCE 7-22 §30.3

GC_p — Roof Field (Zone 1), negative, A = 50 sfGC_p = log-linear interpolation on effective wind arealog₁₀(50) = 1.6990, bounds: [20 sf → -0.9699] to [50 sf → -0.9301]**GC_p = -0.9301****Design Pressure — Roof Field (Zone 1)**p = q_h [(GC_p) - (±GC_{pi})]

Max suction: 31.88 × [(-0.9301) - (+0.18)] = 31.88 × -1.1101 Max positive: 31.88 × [(0.2301) - (-0.18)] = 31.88 × 0.4101

p_{max suction} = -35.39 psf | p_{max positive} = 16.00 psf**GC_p — Roof Edge/Eave (Zone 2), negative, A = 50 sf**GC_p = log-linear interpolation on effective wind arealog₁₀(50) = 1.6990, bounds: [20 sf → -1.5893] to [50 sf → -1.3107]**GC_p = -1.3107****Design Pressure — Roof Edge/Eave (Zone 2)**p = q_h [(GC_p) - (±GC_{pi})]

Max suction: 31.88 × [(-1.3107) - (+0.18)] = 31.88 × -1.4907 Max positive: 31.88 × [(0.2301) - (-0.18)] = 31.88 × 0.4101

p_{max suction} = -47.53 psf | p_{max positive} = 16.00 psf**GC_p — Roof Corner (Zone 3), negative, A = 50 sf**GC_p = log-linear interpolation on effective wind arealog₁₀(50) = 1.6990, bounds: [20 sf → -1.5893] to [50 sf → -1.3107]**GC_p = -1.3107****Design Pressure — Roof Corner (Zone 3)**p = q_h [(GC_p) - (±GC_{pi})]

Max suction: 31.88 × [(-1.3107) - (+0.18)] = 31.88 × -1.4907 Max positive: 31.88 × [(0.2301) - (-0.18)] = 31.88 × 0.4101

p_{max suction} = -47.53 psf | p_{max positive} = 16.00 psf**GC_p — Wall Field (Zone 4), negative, A = 50 sf**

Calc By: MBX

Checked By: JKR

 GC_p = log-linear interpolation on effective wind area $\log_{10}(50) = 1.6990$, bounds: [20 sf $\rightarrow$ -0.9000] to [50 sf $\rightarrow$ -0.8430] $GC_p = -0.8430$ **Design Pressure — Wall Field (Zone 4)** $p = q_h [(GC_p) - (\pm GC_{pi})]$ Max suction: $31.88 \times [(-0.8430) - (+0.18)] = 31.88 \times -1.0230$ Max positive: $31.88 \times [(0.8146) - (-0.18)] = 31.88 \times 0.9946$ $p_{\text{max suction}} = -32.62 \text{ psf}$ | $p_{\text{max positive}} = 31.71 \text{ psf}$ **GC_p — Wall Corner (Zone 5), negative, A = 50 sf** GC_p = log-linear interpolation on effective wind area $\log_{10}(50) = 1.6990$, bounds: [20 sf $\rightarrow$ -1.8000] to [50 sf $\rightarrow$ -1.5722] $GC_p = -1.5722$ **Design Pressure — Wall Corner (Zone 5)** $p = q_h [(GC_p) - (\pm GC_{pi})]$ Max suction: $31.88 \times [(-1.5722) - (+0.18)] = 31.88 \times -1.7522$ Max positive: $31.88 \times [(0.8146) - (-0.18)] = 31.88 \times 0.9946$ $p_{\text{max suction}} = -55.87 \text{ psf}$ | $p_{\text{max positive}} = 31.71 \text{ psf}$ **Roof Pressures**

Zone	Area (sf)	$GC_p (-)$	$GC_p (+)$	Max Suction (psf)	Max Positive (psf)
1 (Field)	10	-1.000	0.300	-37.62	16.00
1 (Field)	20	-0.970	0.270	-36.66	16.00
1 (Field)	50	-0.930	0.230	-35.39	16.00
1 (Field)	100	-0.900	0.200	-34.43	16.00
1 (Field)	200	-0.900	0.200	-34.43	16.00
1 (Field)	500	-0.900	0.200	-34.43	16.00
2 (Edge)	10	-1.800	0.300	-63.13	16.00
2 (Edge)	20	-1.589	0.270	-56.41	16.00
2 (Edge)	50	-1.311	0.230	-47.53	16.00
2 (Edge)	100	-1.100	0.200	-40.81	16.00
2 (Edge)	200	-1.100	0.200	-40.81	16.00
2 (Edge)	500	-1.100	0.200	-40.81	16.00
3 (Corner)	10	-1.800	0.300	-63.13	16.00
3 (Corner)	20	-1.589	0.270	-56.41	16.00
3 (Corner)	50	-1.311	0.230	-47.53	16.00
3 (Corner)	100	-1.100	0.200	-40.81	16.00
3 (Corner)	200	-1.100	0.200	-40.81	16.00
3 (Corner)	500	-1.100	0.200	-40.81	16.00

Wall Pressures

Zone	Area (sf)	$GC_p (-)$	$GC_p (+)$	Max Suction (psf)	Max Positive (psf)
4 (Field)	10	-0.900	0.900	-34.43	34.43
4 (Field)	20	-0.900	0.900	-34.43	34.43
4 (Field)	50	-0.843	0.815	-32.62	31.71

Calc By: MBX

Checked By: JKR

Zone	Area (sf)	$GC_p (-)$	$GC_p (+)$	Max Suction (psf)	Max Positive (psf)
4 (Field)	100	-0.800	0.750	-31.25	29.65
4 (Field)	200	-0.757	0.685	-29.87	27.59
4 (Field)	500	-0.700	0.600	-28.06	24.87
5 (Corner)	10	-1.800	0.900	-63.13	34.43
5 (Corner)	20	-1.800	0.900	-63.13	34.43
5 (Corner)	50	-1.572	0.815	-55.87	31.71
5 (Corner)	100	-1.400	0.750	-50.38	29.65
5 (Corner)	200	-1.228	0.685	-44.88	27.59
5 (Corner)	500	-1.000	0.600	-37.62	24.87

Note: All pressures are subject to a minimum magnitude of 16 psf per §30.2.2. Negative values indicate suction (away from surface); positive values indicate pressure (toward surface).